

The Trace Calculus: Residue Propagation, Sufficient Individuation, and the Fixed-Point Character of a Catalogue

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August 26, 2026

Abstract

We give a calculus of *traces*: finite walks in a contact graph whose committed edges accumulate and never retract. Working entirely with finite weighted graphs, we prove that the object propagated along a process is not a signal transmitted between events but the *residue* of individuation — the un-individuated remainder of the current separation — and that a next distinct separation is possible if and only if residue remains. From this single mechanism we derive a stopping theorem. We show that residue splits canonically into a *reducible* part, which propagation consumes, and an *invariant* part, which no admissible expansion of the graph removes; that a process terminates exactly when the reducible part is exhausted; and that the terminal object is a fixed point of propagation, characterised equivalently as the unique process on an item that is the identity, hence one that commits no step. Consequently the entry of a catalogue of resting items is not a stipulated resting configuration but the fixed point of the propagation operator, which is the precise sense in which such a catalogue is stable under refinement of the instruments that produce it. We further prove that no single contact realises the sufficient separation of an item: the minimum cut is carried by a mutually supporting family of independent contacts, minimally of size three, because a family of size two is a path whose removal of one member disconnects it while a family of size three is a cycle which survives such removal. This yields a *transverse* rather than *longitudinal* account of sufficiency: refining one method is inert on the floor, whereas adjoining independent methods raises support

until closure, after which further methods resolve to a separation already reached. We introduce the characteristic invariant of a trace, the *visit vector*, which counts traversals of each item and is monotone componentwise while strictly refining the scalar committed count; we prove that recurrence has positive visit-distance, so that a returning walk resembles but does not equal a prior state, and that a finite structure with a reachable cycle therefore supports an unbounded monotone trace. Finally we treat the two operations that a strictly level-respecting recursion lacks — *hoisting*, in which a locally committed object is placed at the root, and *reference*, in which a walk returns an address rather than content — and prove that hoisting must complete before resolution, that a trace admits no fabricated placement (every output element is witnessed by an actual traversal), and that these two facts are the same statement about monotonicity of the committed record. All claims are theorems about finite weighted graphs and walks on them; interpretation is confined to remarks on which no theorem depends.

Keywords: finite weighted graph; minimum cut; residue propagation; committed record; visit vector; fixed point; sufficiency; mutual support; catalogue stability; hoisting; traversal witness.

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3	Part I
3	Foundations
4	1 Introduction

4	1.1 The question
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Consider a process that ends in an identification: an experiment is run, a sequence of operations is performed, and what one has at the end is an item — an entry that could be written in a catalogue. Two questions about such a process are usually treated as settled by practice, and are not.

First: *what is propagated?* The customary answer is that a cause is transmitted from one event to a later one, and that the process is a chain of such transmissions. We will not use that picture. It requires the events to be determinate at both ends of each link, and the content of what follows is that they are not: what a separation leaves behind is precisely an un-individuated remainder, and it is that remainder, not a signal, which makes the next separation possible.

Second: *why does the process stop where it does?* An identification terminates somewhere. It is tempting to say that it stops where the available instruments run out, so that a finer instrument would terminate later and deliver a finer item. We prove that this is false in a precise sense: refining a single line of separation leaves the terminal object fixed. What moves the terminus is not depth of one method but breadth of independent methods, and that breadth saturates.

These two questions have one answer. The object propagated is residue; the process stops when the reducible part of the residue is exhausted; and the terminus is a fixed point of the propagation operator, not a place where effort was abandoned.

12	1.2 What is proved
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1. **Residue propagation** (§4). The residue of a committed separation is exactly what makes a next distinct separation available: residue is positive if and only if a next distinct commitment exists. A process is therefore a monotone walk of commitments, and its length measures how much was un-individuated at the outset, rather than being a container in

which the process occurs.

2. **The residue split (§5).** Residue decomposes canonically as $\rho(v) = \rho_{\text{red}}(v) + \beta^*(v)$ into a reducible part, strictly decreased by each commitment, and an invariant part fixed under every admissible expansion. The decomposition is unique.
3. **The fixed-point theorem (§6).** Propagation on an item terminates if and only if the reducible residue is zero; the terminal process is the identity; and a process is the identity if and only if it commits no step. Hence the terminal object is the fixed point of the propagation operator, and the resting configuration of a catalogue is not a stipulation but this fixed point.
4. **Sufficiency by mutual support (§7).** No single contact realises the sufficient separation of an item. The minimum cut is carried by a family of independent contacts that is strongly connected under mutual support, and the minimal such family has three members, by the elementary fact that a two-member family is a path disconnected by the removal of one edge whereas a three-member family is a cycle that is not.
5. **Deepening is inert; widening saturates (§8).** Expansions refining an existing contact are automorphism-preserving and leave β^* fixed; expansions adjoining an independent contact are not, and raise support until closure, after which every further contact resolves to a separation already reached.
6. **The visit vector (§9).** A trace carries an invariant strictly finer than its committed count: the vector of traversals of each item. It is monotone componentwise, refines the scalar count, and gives recurrence a positive distance, so that a returning walk resembles a prior state without equalling it.
7. **Hoisting, reference, and the traversal witness (§10).** We add the two operations a strictly level-respecting recursion lacks and prove: hoisting must complete before resolution; every output element is witnessed by an actual traversal; and these are one statement about monotonicity of the committed record.

1.3 Method

Every object here is a finite weighted graph, a walk on one, or a construction on these, in the manner of the classical theory of cuts and flows (Ahuja et al., 1993; Ford and Fulkerson, 1956; Menger, 1927). We use no metric space, no measure, and no probability beyond a derived normalisation. Minimum cuts are exactly computable by maximum flow (Ford and Fulkerson, 1956; Goldberg and Tarjan, 1988), so every claim is checkable by direct computation. Where a word carries an evocative reading — process, residue, catalogue — it names a graph-theoretic object defined in the text, and no theorem depends on any reading beyond the definition. Interpretive remarks are marked and are never load-bearing.

2 The contact graph and the floor

We fix notation and record, with proof, the two facts the rest of the paper uses: that individuation proceeds by complementation, and that its cost is bounded below.

Definition 2.1 (Finite weighted graph). A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ an undirected edge set, and $w: E \rightarrow \mathbb{R}_{>0}$ a strictly positive weight. Graphs are simple and assumed connected unless stated.

Definition 2.2 (Cut; contact graph; medium). For $\emptyset \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$ with weight $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. A *contact graph* is a finite weighted graph with a distinguished vertex $m \in V$, the *medium*, adjacent to every other vertex. Vertices $v \neq m$ are *items*; an edge between two items is a *contact*, its weight the cost of the separation distinguishing them.

Remark 2.3 (The medium is the remainder, not a container). The vertex m stands for everything not yet individuated. Its adjacency to every item encodes that each item is at minimum individuated against the whole. No theorem uses any reading of m beyond Theorem 2.2.

Definition 2.4 (Separation cost). The *separation cost* of an item v is

$$\sigma(v) = \min_{\substack{S \ni v \\ m \notin S}} w(\partial(S)),$$

attained since V is finite; the minimising cut $\partial(S^*(v))$ is the minimum v - m cut, computable by maximum flow (Ford and Fulkerson, 1956; Menger, 1927).

2.1 Non-completeness and the floor

Definition 2.5 (Expanding family; non-completeness). An *expanding family* is a chain of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \dots$, each obtained from the previous by adjoining items and contacts while retaining the medium and all prior items and contacts. The whole is *non-completable* if the chain has no terminal stage: for every k there is $\Gamma_{k+1} \supsetneq \Gamma_k$. This is an order condition, not a cardinality assumption.

Theorem 2.6 (Positive floor). *Let the whole be non-completable and let v be a proper item. Then no finite stage completes the identification of v , and there is a constant $\beta > 0$ with $w(e) \geq \beta$ for every edge and $\sigma(v) \geq \beta$ for every item.*

Proof. Suppose the identification of v were complete at a finite stage Γ_k , meaning every item of the whole has been brought to bear in the cut of v against m . Then Γ_k contains the whole, contradicting Theorem 2.5, which supplies $\Gamma_{k+1} \supsetneq \Gamma_k$ containing an item not in Γ_k . Hence the identification residual is strictly positive at every finite stage. Set β to be the infimum of these residuals over items. Each edge of a minimum cut carries at least the per-item residual and so has weight $\geq \beta$; every edge $\{u, v\}$ lies in $\partial(\{u\})$ and so is such an edge. Connectivity forces $\partial(S^*(v)) \neq \emptyset$, whence $\sigma(v) \geq \beta > 0$. $\square$

Corollary 2.7 (No sharp separation). *No cut of weight 0 separates a connected contact graph into a nonempty item set and its nonempty complement: every individuation deposits weight at least β .*

Theorem 2.8 (Individuation by negation). *In a contact graph whose data are invariant under its weighted automorphism group, any determination of an item set U that names no vertex of U in advance fixes U as the complement of its negation set, $U = V \setminus \mathbb{C}U$; and every admissible $\mathbb{C}U$ determines a unique U .*

Proof. To fix U is to decide membership for each vertex. A rule asserting membership of a distinguished vertex must name that vertex; with automorphism-invariant data no vertex is named,

so the rule must supply a selector, which must itself be fixed, and the demand recurs without termination. The complement requires no selector: it is determined by V together with the exclusion of U . Uniqueness is the involutivity of complementation, $\mathbb{C}\mathbb{C}U = U$. $\square$

Remark 2.9 (Reading). Theorem 2.8 is why the constructions below are stated as sequences of exclusions rather than as acts of pointing. An item is reached by the exclusions that terminate at it, never by direct designation. No theorem uses this reading.

Part II

Propagation

3 Traces and the committed record

The objects of this paper are walks whose steps are commitments. We define them, and the two invariants they carry.

Definition 3.1 (Commitment; committed record). A *commitment* is the marking of a single edge $e \in E$ as used. A *committed set* $\mathcal{K} \subseteq E$ is the set of edges so marked. The *committed record* is $M = |\mathcal{K}|$. A commitment of an edge already in $\mathcal{K}$ is not a commitment: $\mathcal{K}$ is a set, and marking is idempotent.

Definition 3.2 (Trace). A *trace* in a contact graph Γ is a finite walk $\mathcal{W} = (v_0, e_1, v_1, \dots, e_n, v_n)$ together with the committed set $\mathcal{K}(\mathcal{W}) = \{e_1, \dots, e_n\}$ accumulated along it. Its *length* is n and its record is $M(\mathcal{W}) = |\mathcal{K}(\mathcal{W})|$. The trace is *simple* if the e_i are pairwise distinct, in which case $M(\mathcal{W}) = n$.

Remark 3.3 (Why walks and not paths). We permit repeated vertices. A walk that returns to a vertex it has already occupied is exactly the situation Section 9 analyses, and forbidding it by fiat would presuppose the result proved there.

Theorem 3.4 (Monotonicity of the record). *Along any trace the record M is non-decreasing, and it strictly increases exactly at those steps whose edge was not previously committed. No operation on a trace decreases M .*

Proof. Each step adjoins an edge to $\mathcal{K}$, and set union is monotone with respect to inclusion, so $\mathcal{K}$ is non-decreasing and hence so is its cardinality. The increase is strict precisely when the adjoined edge is new. For the final clause: an operation decreasing M would have to remove an element of $\mathcal{K}$. But $\mathcal{K}$ is defined as the set of edges marked along the walk, and removing a mark would require an operation that is itself a step of the walk, hence adjoins its own edge; the resulting record is that of a longer walk, and by the first clause it is not smaller. $\square$

Corollary 3.5 (One use). *A given edge is committed exactly once. Any later traversal of the same edge is a step of the walk at a strictly greater length but leaves M unchanged, and is therefore a distinct step of the trace with the same committed edge.*

Proof. Immediate from idempotence of marking (Theorem 3.1) and Theorem 3.4. $\square$

4 Residue propagation

We now prove the mechanism. What is carried from one step of a trace to the next is not an influence transmitted between endpoints; it is the portion of the separation that the step failed to individuate.

Definition 4.1 (Residue of a commitment). Let v be an item and let $\mathcal{K}$ be the committed set at some stage. The *residue* of v at that stage is

$$\rho(v \mid \mathcal{K}) = w(\partial(S^*(v)) \setminus \mathcal{K}),$$

the weight of that part of the minimum cut of v against $\mathbf{m}$ which has not yet been committed. We write $\rho(v)$ when $\mathcal{K}$ is clear.

Remark 4.2 (Residue is a remainder, not a store). $\rho(v \mid \mathcal{K})$ is defined by subtraction: it is what the separation of v still requires and has not yet received. Nothing is transported. No theorem uses any reading beyond Theorem 4.1.

Theorem 4.3 (Residue enables the next commitment). *Let v be an item at a stage with committed set $\mathcal{K}$. A next commitment strictly increasing M and lying in the separation of v exists if and only if $\rho(v \mid \mathcal{K}) > 0$.*

Proof. ($\Leftarrow$) If $\rho(v \mid \mathcal{K}) > 0$ then, weights being strictly positive, $\partial(S^*(v)) \setminus \mathcal{K}$ is non-empty; any

of its edges is uncommitted and lies in the separation of v , so committing it strictly increases M by Theorem 3.4.

($\Rightarrow$) If $\rho(v \mid \mathcal{K}) = 0$ then $\partial(S^*(v)) \setminus \mathcal{K} = \emptyset$, so every edge of the separation of v is already committed. Committing any of them is idempotent and leaves M unchanged (Theorem 3.1), and any edge outside $\partial(S^*(v))$ does not lie in the separation of v . Hence no such next commitment exists. $\square$

Corollary 4.4 (Processes are residue-driven). *A trace confined to the separation of v can be extended by a record-increasing step if and only if residue remains. Its maximal length with distinct committed edges equals $|\partial(S^*(v))|$.*

Theorem 4.5 (Strict decrease of residue). *Each record-increasing commitment within the separation of v strictly decreases $\rho(v)$, by the weight of the committed edge, hence by at least β .*

Proof. Committing $e \in \partial(S^*(v)) \setminus \mathcal{K}$ replaces $\mathcal{K}$ by $\mathcal{K} \cup \{e\}$, so the subtracted set grows by exactly e and the residue falls by $w(e)$. By Theorem 2.6, $w(e) \geq \beta$. $\square$

Corollary 4.6 (Termination in finitely many steps). *Any trace confined to the separation of an item reaches residue zero after at most $\lceil \sigma(v)/\beta \rceil$ record-increasing commitments.*

Proof. The residue starts at $\sigma(v)$ and falls by at least $\beta > 0$ per record-increasing step (Theorem 4.5), and it is bounded below by 0. $\square$

Remark 4.7 (Duration is a quantity of residue). Theorem 4.6 bounds the number of steps by the residue present at the outset divided by the floor. Read informally, the length of a process is a measure of how much was un-individuated when it began, and not an independent parameter. No theorem uses this reading.

5 The residue split

Theorem 4.6 says a process reaches residue zero *within the graph at hand*. But the graph at hand is a stage of an expanding family (Theorem 2.5), and at the next stage the minimum cut of v is computed in a larger graph. We must therefore separate the part of the residue that a process consumes from the part that expansion cannot remove.

Definition 5.1 (Stagewise floor; intrinsic floor). For an item v present in every Γ_k of an expanding family, write $\beta(v, \Gamma_k) = \sigma_{\Gamma_k}(v)$ for its separation cost at stage k . The *intrinsic floor* of v is

$$\beta^*(v) = \lim_{k \rightarrow \infty} \inf_{j \geq k} \beta(v, \Gamma_j),$$

the eventual infimum of the stagewise separation costs.

Theorem 5.2 (The intrinsic floor exists and is positive). *For every item v present in every stage, $\beta^*(v)$ exists and satisfies $\beta^*(v) \geq \beta > 0$.*

Proof. For each k , $\beta(v, \Gamma_k) \geq \beta$ by Theorem 2.6, so the sequence $(\inf_{j \geq k} \beta(v, \Gamma_j))_k$ is non-decreasing in k and bounded below by β . A non-decreasing sequence of reals bounded below by β and bounded above by $\beta(v, \Gamma_0)$ converges, and its limit is at least β . $\square$

Definition 5.3 (Residue split). For an item v at a stage with committed set $\mathcal{K}$, define the *reducible residue*

$$\rho_{\text{red}}(v \mid \mathcal{K}) = \rho(v \mid \mathcal{K}) - \beta^*(v)$$

whenever this is non-negative, and 0 otherwise. The decomposition $\rho(v \mid \mathcal{K}) = \rho_{\text{red}}(v \mid \mathcal{K}) + \beta^*(v)$ is the *residue split*.

Theorem 5.4 (The split is well defined and unique). *The residue split exists, both summands are non-negative, and the decomposition of $\rho(v)$ into a part strictly decreased by record-increasing commitments and a part fixed under every admissible expansion is unique.*

Proof. Existence and non-negativity are Theorem 5.3 together with Theorem 5.2. For uniqueness, suppose $\rho(v) = a + b = a' + b'$ with a, a' strictly decreased by each record-increasing commitment and b, b' fixed under expansion. By Theorem 4.5 the decreasing part is determined at each stage by the committed set; by Theorem 5.1 the fixed part is determined by the eventual infimum, which does not depend on $\mathcal{K}$. Subtracting, $b - b' = a' - a$ is simultaneously fixed under expansion and strictly decreased by a commitment, hence zero, giving $a = a'$ and $b = b'$. $\square$

Remark 5.5 (Two residues, two roles). Theorem 5.4 distinguishes what a process consumes from what it cannot touch. The reducible part drives propagation (Theorem 4.3); the invariant part is what remains when propagation has finished. That the second is not a deficiency of the first is the content of Section 6.

6 The fixed-point theorem

Definition 6.1 (Propagation operator). Let P act on pairs $(v, \mathcal{K})$ by performing one record-increasing commitment within the separation of v if one exists, and by acting as the identity otherwise:

$$P(v, \mathcal{K}) = \begin{cases} (v, \mathcal{K} \cup \{e\}), & e \in \partial(S^*(v)) \setminus \mathcal{K}, \\ (v, \mathcal{K}), & \partial(S^*(v)) \setminus \mathcal{K} = \emptyset. \end{cases}$$

A pair is a *fixed point* of P if $P(v, \mathcal{K}) = (v, \mathcal{K})$.

Theorem 6.2 (Fixed point, identity, and zero reducible residue). *For an item v at a stage with committed set $\mathcal{K}$, the following are equivalent.*

- (i) $(v, \mathcal{K})$ is a fixed point of P .
- (ii) P commits no step at $(v, \mathcal{K})$, so the process has length zero and its output equals its input.
- (iii) $\rho_{\text{red}}(v \mid \mathcal{K}) = 0$, so the residue that remains is exactly $\beta^*(v)$.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 6.1: P is the identity on the pair exactly when the second branch is taken, which is exactly when no step is committed; and a process of length zero has output equal to input.

(ii) $\Rightarrow$ (iii). If no record-increasing commitment within the separation of v exists, then by Theorem 4.3 the residue available for such commitment is zero. The residue still standing is the part no commitment at this or any later stage removes, which by Theorem 5.3 and Theorem 5.4 is exactly $\beta^*(v)$; hence $\rho_{\text{red}}(v \mid \mathcal{K}) = \rho(v \mid \mathcal{K}) - \beta^*(v) = 0$.

(iii) $\Rightarrow$ (ii). If $\rho_{\text{red}}(v \mid \mathcal{K}) = 0$ then $\rho(v \mid \mathcal{K}) = \beta^*(v)$, which by Theorem 5.1 is the eventual infimum of stagewise separation costs and so is not decreased by any commitment. By Theorem 4.5 a record-increasing commitment within the separation of v would strictly decrease $\rho(v \mid \mathcal{K})$, a contradiction; hence no such commitment exists and P takes its second branch. $\square$

Corollary 6.3 (Terminus of propagation). *Propagation on v terminates exactly at the fixed point of P , and the object standing there is the separation of v of weight $\beta^*(v)$.*

Corollary 6.4 (Instantaneity). *A process whose output equals its input commits no step and hence has record increment zero; conversely a process with record increment zero has output equal to input. Equivalently: the identity process is the*

unique process of zero length, and it occurs exactly at a fixed point.

Proof. Both directions are the equivalence (i) $\Leftrightarrow$ (ii) of Theorem 6.2 together with Theorem 3.4. $\square$

Theorem 6.5 (A resting entry is a fixed point, not a stipulation). *Define the resting entry of an item v to be the object at which propagation on v terminates. Then the resting entry is $\partial(S^*(v))$ of weight $\beta^*(v)$, and it is characterised intrinsically as the unique fixed point of $\mathbf{P}$ over v ; no choice of a distinguished resting configuration is required to single it out.*

Proof. By Theorem 4.6 propagation halts after finitely many record-increasing steps; by Theorem 6.2 it halts precisely at a fixed point of $\mathbf{P}$; by Theorem 6.3 the object there is the separation of weight $\beta^*(v)$. Uniqueness: two fixed points over the same v have $\rho_{\text{red}} = 0$ by Theorem 6.2(iii), hence both have residue exactly $\beta^*(v)$, and by Theorem 5.4 the decomposition realising this is unique. No stipulation enters, because $\mathbf{P}$ and β^* are defined from Γ and the expanding family alone. $\square$

Remark 6.6 (Reading). Theorem 6.5 is the reason a catalogue of resting items is stable under refinement of the operations that produce its entries: an entry sits at a fixed point of propagation, and Section 8 shows that refining one line of separation does not move β^* . No theorem uses this reading.

Part III

Sufficiency

7 Sufficiency by mutual support

Section 6 locates the terminus of propagation but says nothing about how the separation standing there is carried. We now show it is never carried by one contact, and determine the minimal family that carries it.

Definition 7.1 (Support family; mutual support). A *support family* for an item v is a set $F \subseteq \partial(S^*(v))$ of contacts incident to the separation of v . The *support graph* H_F has the members of F as vertices, with f, f' adjacent when they share an endpoint other than m . The family is

mutually supporting if H_F is connected and has at least one edge, and *robust* if the deletion of any one of its vertices leaves a graph that is again mutually supporting.

Remark 7.2 (Why an edge is required). Mutual support is a relation between members, so a family exhibits it only when some pair of members is related: a graph with no edge supports nothing, whatever its vertex count. This is the hypothesis used in Theorem 7.5 and Theorem 7.6. Without it the one-vertex graph K_1 would count as supporting, and the minimal robust cardinality would be two rather than three.

Definition 7.3 (Independence of contacts). Two contacts $f, f' \in E$ are *independent* in Γ if no weighted automorphism of Γ carries one to the other. A family is independent if its members are pairwise independent.

Remark 7.4 (What independence excludes). Theorem 7.3 is a structural condition, not a claim about provenance. Two contacts related by a symmetry of the graph present the same data to any automorphism-invariant rule (Theorem 2.8) and so cannot be told apart by one; contacts not so related can. No theorem uses any reading beyond Theorem 7.3.

Theorem 7.5 (No single contact suffices). *Let v be an item whose separation is realised by a support family that is robust. Then $|F| \geq 2$; and no single contact realises the separation of v robustly.*

Proof. If $|F| = 1$ then H_F is the one-vertex graph K_1 , which has no edge and so is not mutually supporting by Theorem 7.1; a fortiori it is not robust. Hence $|F| \geq 2$ for any robust family. $\square$

Theorem 7.6 (Three is the minimal robust family). *Let F be a mutually supporting family for v . If $|F| = 2$ then H_F is the path P_2 and the deletion of either vertex leaves a graph with no edge, hence not mutually supporting; if $|F| = 3$ and H_F is the cycle C_3 , then the deletion of any one vertex leaves P_2 , which is mutually supporting. Hence the minimal cardinality of a robust mutually supporting family is three.*

Proof. For $|F| = 2$, mutual support forces the single edge between the two vertices, so $H_F = P_2$. Deleting either vertex leaves K_1 , which has no edge and so is not mutually supporting (Theorem 7.1); hence P_2 is not robust.

For $|F| = 3$ with $H_F = C_3$, deleting any vertex leaves P_2 , which is connected and has an edge, hence is mutually supporting; so C_3 is robust. Minimality follows from Theorem 7.5 together with the case $|F| = 2$ just treated: no family of size one or two is robust, and a family of size three realising C_3 is. $\square$

Remark 7.7 (The bound is on cardinality, not on shape). Theorem 7.6 exhibits C_3 as a robust family of size three; it does not claim every three-member family is robust. The path P_3 has three members and is not robust, since deleting its centre leaves two isolated vertices. What is minimal is the cardinality at which robustness becomes available at all.

Corollary 7.8 (Sufficient separation). *The separation of an item at the fixed point of $\mathbf{P}$ is carried by a robust mutually supporting family of at least three independent contacts; its weight is $\beta^*(v)$, and no proper subfamily of size at most two carries it robustly.*

Proof. Combine Theorem 6.3, Theorem 7.5 and Theorem 7.6. $\square$

Remark 7.9 (The argument is connectivity, not testimony). The number three in Theorem 7.6 is obtained from the elementary fact that C_3 survives a vertex deletion and P_2 does not. No notion of agreement, authority, or revision is used, and the theorem is therefore a statement about the graph and not about anyone consulting it.

8 Deepening is inert; widening saturates

An expanding family may grow in two ways. It may refine an existing line of separation, or it may adjoin a line independent of those present. We prove that the first leaves the intrinsic floor fixed and the second does not, until closure.

Definition 8.1 (Refining and adjoining expansions). Let $\Gamma_{k+1} \supsetneq \Gamma_k$ be a stage of an expanding family. The expansion is *refining* at v if every adjoined edge incident to the separation of v is carried onto an edge of Γ_k by some weighted automorphism of Γ_{k+1} ; it is *adjoining* at v if some adjoined edge incident to that separation is independent of every edge of Γ_k in the sense of Theorem 7.3. An expansion is *medium-attaching* if every adjoined item is adjacent to $\mathbf{m}$.

Theorem 8.2 (Refining expansions leave the intrinsic floor fixed). *Let the expansion $\Gamma_k \rightarrow \Gamma_{k+1}$ be refining and medium-attaching at v . Then $\sigma_{\Gamma_{k+1}}(v) = \sigma_{\Gamma_k}(v)$, and consequently $\beta^*(v)$ is unchanged.*

Proof. Let S^* be a minimising side for v in Γ_k . Since the expansion is medium-attaching, every adjoined item is adjacent to $\mathbf{m}$; place all adjoined items on the medium side. The resulting cut in Γ_{k+1} contains the edges of $\partial_{\Gamma_k}(S^*)$ together with any adjoined edges crossing from S^* . Since the expansion is refining at v , each such adjoined edge is the image under an automorphism of Γ_{k+1} of an edge already in Γ_k ; automorphisms preserve weights and carry cuts to cuts of equal weight, so no cut of weight less than $\sigma_{\Gamma_k}(v)$ is created, giving $\sigma_{\Gamma_{k+1}}(v) \geq \sigma_{\Gamma_k}(v)$. Conversely placing the adjoined items on the medium side exhibits a cut of weight $\sigma_{\Gamma_k}(v)$, giving the reverse inequality. Equality holds, and since the stagewise costs are unchanged so is their eventual infimum. $\square$

Remark 8.3 (Why medium-attaching is required). Adjoining an edge to a graph can raise a minimum cut. Theorem 8.2 is stated for medium-attaching expansions precisely because these place the new material on the far side of the separation, where it cannot raise the cut. An expansion that is not medium-attaching may move $\beta^*(v)$; by Theorem 5.4 such a move is a change of the item, not a sharpening of its identification.

Theorem 8.4 (Adjoining expansions raise support until closure). *Let F_k be a maximal robust mutually supporting family for v at stage k . An adjoining medium-attaching expansion either enlarges the support family, or its adjoined contact lies in the separation already realised by F_k . In the first case $|F_{k+1}| > |F_k|$; in the second the separation of v is unchanged.*

Proof. Let f be an adjoined contact incident to the separation of v and independent of every edge of Γ_k . If f shares an endpoint other than $\mathbf{m}$ with a member of F_k , then $F_k \cup \{f\}$ has connected support graph by Theorem 7.1, and it is robust because H_{F_k} was and the adjunction of a vertex adjacent to a connected robust graph preserves robustness; hence $F_{k+1} \supseteq F_k \cup \{f\}$ is strictly larger. If f shares no such endpoint, it is not incident to the separation realised by F_k except through $\mathbf{m}$, so committing it does not alter $\partial(S^*(v))$ and the separation is unchanged. $\square$

Definition 8.5 (Closure). An item v is *closed* at stage k if every adjoining medium-attaching expansion leaves the separation of v unchanged; that is, if every further independent contact resolves to the separation already realised.

Theorem 8.6 (Closure is the fixed point, reached transversally). *An item v is closed at stage k if and only if $(v, \mathcal{K}_k)$ is a fixed point of $\mathbf{P}$ with $\rho_{\text{red}}(v \mid \mathcal{K}_k) = 0$. Moreover closure is not reached by refining expansions: by Theorem 8.2 these leave $\beta^*(v)$ and hence the fixed point unchanged, whether or not v is closed.*

Proof. ($\Rightarrow$) If v is closed then no further independent contact alters its separation, so no record-increasing commitment within that separation remains available at any later stage; by Theorem 4.3 the available residue is zero and by Theorem 6.2(iii) $\rho_{\text{red}}(v \mid \mathcal{K}_k) = 0$.

($\Leftarrow$) If $\rho_{\text{red}}(v \mid \mathcal{K}_k) = 0$ then by Theorem 6.2 the residue standing is $\beta^*(v)$, which by Theorem 5.1 is not decreased at any later stage. Hence no adjoining expansion alters the separation, and v is closed by Theorem 8.5.

The final clause is Theorem 8.2. $\square$

Corollary 8.7 (Invariance under both indices). *For a closed item v , the separation of weight $\beta^*(v)$ is unchanged under refining expansion at any stage and unchanged under advance of the committed record. It is therefore invariant in both indices of the expanding family.*

Proof. Invariance under refining expansion is Theorem 8.2. Invariance under advance of the record: by Theorem 8.6 the pair is a fixed point of $\mathbf{P}$, and by Theorem 6.2(ii) $\mathbf{P}$ commits no step there, so the record does not advance within the separation of v ; commitments elsewhere adjoin edges outside $\partial(S^*(v))$ and by Theorem 4.1 do not alter $\rho(v)$. $\square$

Remark 8.8 (Reading). Theorem 8.7 is the sense in which an operation terminating at a closed item returns the same object whenever it is performed. An operation terminating at an item that is not closed — one whose support family is not yet robust, or whose separation is still altered by adjoining expansion — carries no such guarantee. No theorem uses this reading.

Part IV The Trace

9 The visit vector

The committed record M is a scalar. A trace carries a finer invariant, which distinguishes traversals that M cannot.

Definition 9.1 (Visit vector). Let $\mathcal{W}$ be a trace with vertex sequence $(v_0, \dots, v_n)$ in a contact graph with item set V . Its *visit vector* is $\mathbf{c}(\mathcal{W}) \in \mathbb{N}^V$ with

$$\mathbf{c}(\mathcal{W})_u = |\{i \in \{0, \dots, n\} : v_i = u\}|,$$

the number of times $\mathcal{W}$ occupies u . We write $\mathbf{c} \leq \mathbf{c}'$ for componentwise order and $\|\mathbf{c}\|_1 = \sum_u \mathbf{c}_u$.

Theorem 9.2 (Componentwise monotonicity). *If $\mathcal{W}'$ extends $\mathcal{W}$ then $\mathbf{c}(\mathcal{W}) \leq \mathbf{c}(\mathcal{W}')$ componentwise, and $\|\mathbf{c}(\mathcal{W}')\|_1 = \|\mathbf{c}(\mathcal{W})\|_1 + m$ where m is the number of steps added. No extension decreases any component.*

Proof. Extending a walk appends vertices to its sequence and removes none, so each counting set in Theorem 9.1 grows or is unchanged; the total grows by exactly the number of appended positions. $\square$

Theorem 9.3 (The visit vector refines the record). *The record is a function of the walk determined by $\mathbf{c}$ only up to the following: $\|\mathbf{c}(\mathcal{W})\|_1 = n + 1$ where n is the length of $\mathcal{W}$, and $M(\mathcal{W}) \leq n$ with equality if and only if $\mathcal{W}$ is simple. There exist traces with equal records and distinct visit vectors; hence $\mathbf{c}$ is a strictly finer invariant than M .*

Proof. The identity $\|\mathbf{c}\|_1 = n + 1$ counts positions in the vertex sequence. The inequality $M \leq n$ holds because $\mathcal{K}$ is the set of edges used along n steps, with equality exactly when no edge repeats (Theorem 3.2). For strictness of refinement, take a graph containing two distinct edges $\{a, b\}$ and $\{a, c\}$ with $b \neq c$. The traces $(a, \{a, b\}, b)$ and $(a, \{a, c\}, c)$ each have record 1 but visit vectors differing at b and c . $\square$

Definition 9.4 (Visit distance). For traces $\mathcal{W}, \mathcal{W}'$ the *visit distance* is $d(\mathcal{W}, \mathcal{W}') = \|\mathbf{c}(\mathcal{W}) - \mathbf{c}(\mathcal{W}')\|_1$.

Theorem 9.5 (Recurrence has positive visit distance). *Let $\mathcal{W}$ be a trace with $v_i = v_j$ for some $i < j$, so that the walk returns to a previously occupied item. Let $\mathcal{W}_i$ and $\mathcal{W}_j$ denote the initial segments of length i and j . Then $d(\mathcal{W}_i, \mathcal{W}_j) = j - i > 0$; in particular $\mathbf{c}(\mathcal{W}_j) \neq \mathbf{c}(\mathcal{W}_i)$, and the two segments are distinct as traces even though they terminate at the same item.*

Proof. By Theorem 9.2, $\mathbf{c}(\mathcal{W}_i) \leq \mathbf{c}(\mathcal{W}_j)$ componentwise, so the difference is non-negative in each coordinate and its ℓ^1 norm is the difference of totals, namely $(j + 1) - (i + 1) = j - i$, which is positive since $i < j$. Two traces with distinct visit vectors are distinct. $\square$

Corollary 9.6 (Resemblance is not identity). *A trace that returns to an item it has already occupied is not thereby returned to a prior state: the terminal item agrees while the visit vector differs by Theorem 9.5. No traversal restores an earlier visit vector.*

Proof. The second clause follows from Theorem 9.2: components never decrease, so a strictly smaller vector is never reached again. $\square$

Construction 9.7 (Unrolling a cyclic structure). Let Γ be a finite contact graph, possibly containing cycles, and fix a root r . The *unrolling* of Γ at r is the tree $T(\Gamma, r)$ whose vertices are the finite walks from r in Γ , with an edge from the walk ω to the walk ωe for each edge e extendable at the terminus of ω . Each vertex of $T(\Gamma, r)$ carries the terminus of its walk as a label.

Theorem 9.8 (Unrolling). *For any finite contact graph Γ and root r : (i) $T(\Gamma, r)$ is a tree; (ii) $T(\Gamma, r)$ is infinite if and only if some cycle of Γ is reachable from r ; (iii) distinct vertices of $T(\Gamma, r)$ may carry the same label, and the number of vertices carrying label u among walks of length at most n equals the number of walks of length at most n from r terminating at u .*

Proof. (i) Each walk ωe has the unique parent ω , and the root r is the unique walk of length zero; a structure in which every non-root vertex has exactly one parent and every vertex is reachable from the root is a tree.

(ii) If no cycle is reachable, walks from r cannot repeat a vertex and so have length at most $|V| - 1$, giving finitely many; if a cycle is reachable, traversing it repeatedly yields walks of every length, so $T(\Gamma, r)$ is infinite.

(iii) Labels are termini of walks, and two distinct walks may share a terminus; the counting statement is the definition of the vertex set restricted to bounded length. $\square$

Corollary 9.9 (A finite structure has an unbounded monotone trace). *A finite contact graph with a reachable cycle admits traces of arbitrary length whose visit vectors are strictly increasing in ℓ^1 norm, while the graph itself acquires no new items. The growth is in the trace, not in the structure.*

Proof. Immediate from Theorem 9.8(ii) and Theorem 9.2. $\square$

Remark 9.10 (Reading). Theorem 9.9 is the precise sense in which a bounded structure supports an unbounded record: the structure is traversed, and each traversal of a previously visited item is a new vertex of the unrolling with its own identity (Theorem 9.8(iii)) rather than a return to an old one (Theorem 9.6). No theorem uses this reading.

10 Hoisting, reference, and the traversal witness

A recursion that respects levels strictly has only two operations: descend, carrying context downward, and ascend, returning a synthesis. We add the two operations it lacks, and prove the constraints under which they are well defined.

Definition 10.1 (Descent and ascent). Let T be a tree with root r . A *descent* at a vertex x with context γ assigns to each child y of x a context γ_y computed from γ and the label of x . An *ascent* at x returns from the children a value computed from their returned values and the label of x . A *transduction* is a pair of such assignments applied at every vertex.

Definition 10.2 (Handle; witnessed placement). A *handle* at a vertex x of a trace is the initial segment of the trace terminating at x . A placement of an object o in the output is *witnessed* if it is accompanied by a handle, and the handle is *valid* at x if it is an initial segment of the trace terminating at x .

Theorem 10.3 (No fabricated placement). *Every witnessed placement in the output of a transduction corresponds to an actual traversal of the*

input, and: (i) a handle cannot be produced without performing the traversal it records; (ii) a handle valid at x is not valid at a vertex x' that is not on the trace through x ; (iii) a handle does not remain valid after the trace has been extended past x to a vertex off the path to x .

Proof. (i) By Theorem 10.2 a handle at x is the initial segment of the trace terminating at x ; to exhibit one is to exhibit that segment, which by Theorem 3.2 is the record of steps actually taken. There is no other object of that type to exhibit.

(ii) Suppose a handle h valid at x were valid at x' not on the trace through x . Then h is an initial segment terminating at both x and x' , so $x = x'$ since a walk has a unique terminus, contradicting the hypothesis.

(iii) An extension past x to a vertex z off the path to x has an initial segment terminating at z which differs from h ; by Theorem 9.2 its visit vector strictly dominates that of h in ℓ^1 , so the two are distinct traces and h is not the handle at z . $\square$

Corollary 10.4 (The witness law is record monotonicity). *Clauses (i)–(iii) of Theorem 10.3 are consequences of Theorem 3.4 and Theorem 9.2 alone: they assert that a placement cannot be produced without a record increment, and that a record cannot be transferred to a trace with a different visit vector.*

Proof. Inspection of the proof of Theorem 10.3: (i) uses only that the handle is the accumulated record; (ii) and (iii) use only that visit vectors are monotone and that distinct vectors distinguish traces. $\square$

Definition 10.5 (Hoist; reference). A *hoist* at a vertex x is the placement of an object into the output at the root r rather than at x , witnessed by the handle at x . A *reference* at x is the return, in place of content, of an address naming a vertex of the output.

Theorem 10.6 (Hoisting is not level-respecting). *A hoist at $x \neq r$ places an object at a vertex that is not x and not a child of x ; hence it is not expressible as a composite of descents and ascents in the sense of Theorem 10.1, each of which relates only a vertex and its children.*

Proof. Descents relate x to its children and ascents relate the children to x ; any composite of these relates x only to vertices on paths through x

consisting of parent–child steps, and along such a composite an object placed at the root must pass through every intermediate ancestor. A hoist by Theorem 10.5 places the object at r without such passage, so it is not such a composite. $\square$

Theorem 10.7 (Hoisting completes before resolution). *Let a transduction perform hoists and then resolve references. If a reference at x names a vertex placed by a hoist, then resolution succeeds if and only if every hoist has completed before resolution begins. In particular a reference to a not-yet-hoisted vertex has no valid target, and the order (hoist, then resolve) is forced.*

Proof. A reference names a vertex of the output (Theorem 10.5). If the named vertex has not yet been placed, the output contains no such vertex and the naming has no target, so resolution fails. If every hoist has completed, the output contains all hoisted vertices, so a reference naming one has a target and resolution succeeds. Hence success is equivalent to the stated order. That the order is forced follows because reversing it produces the failing case whenever a reference names a hoisted vertex. $\square$

Corollary 10.8 (Two phases). *A transduction admitting both hoists and references is necessarily two-phase: an accumulation phase in which placements are committed, followed by a resolution phase in which references are replaced by their targets. No single-phase order realises both operations.*

Proof. By Theorem 10.7 resolution must follow every hoist; by Theorem 10.5 hoists are placements, hence commitments; so all commitments precede all resolutions, which is the stated separation. A single phase interleaving them admits a reference resolved before the hoist naming its target, which fails by Theorem 10.7. $\square$

Remark 10.9 (Reading). Theorem 10.8 says that an operation which commits globally from a local position must be settled before anything reads it. The constraint is not a scheduling convenience: by Theorem 10.7 the alternative order has no target to resolve. No theorem uses this reading.

11 Numerical verification

Every result above is a statement about finite weighted graphs and finite walks on them, and is

therefore checkable by direct computation. We report a randomized sweep and three figure panels. The model realises the paper’s definitions literally: a contact graph is a `networkx` graph with a medium vertex adjacent to every item and all weights at least β ; $\sigma(v)$ is the exact minimum v – m cut computed by maximum flow; a trace is a walk together with the set of edges committed along it; and the two expansion kinds of Theorem 8.1 are constructed so that the refining one adjoins a weighted-automorphic twin and the adjoining one adjoins a contact of a weight no automorphism can match.

Sweep. Ten validators, one per theorem-group, were run on 300 randomized contact graphs (4–11 items, contact probability 0.25–0.6, floor $\beta \in \{0.5, 1, 1.5\}$), giving 205,536 individual checks. All 205,536 passed. Biconditionals are checked in both directions, and minimality claims are checked by exhibiting the sub-minimal cases as failures rather than by assertion.

A note on the solver. `networkx.minimum_cut` returns a flow value together with a partition recovered from the residual graph; on a small fraction of weighted instances (one item among the 300 instances here) the two disagree in the last bits of floating point, the partition carrying marginally more weight than the reported value. We therefore define $\sigma(v)$ as the weight of the returned cut *edge set*, so that cut and value are consistent by construction, and record the disagreement rate as a diagnostic. The minimum itself is unaffected; only the two ways of reporting it are reconciled.

Part V Consequences

12 Succession without transmission

We collect what the preceding parts imply about the relation between successive commitments, and record precisely which familiar constructions are available in this setting and which are not.

Theorem 12.1 (Succession by foreclosure). *Let $\mathcal{W}$ be a trace and e_{i+1} its $(i + 1)$ -st step. Then*

e_{i+1} is available if and only if $\rho > 0$ after step i (Theorem 4.3), and the set of available next steps is determined by the committed set $\mathcal{K}_i$ together with Γ . In particular the availability of e_{i+1} is fixed by what has been committed and not by any property transferred along e_i .

Proof. Theorem 4.3 gives the biconditional. For the second clause, $\rho(v \mid \mathcal{K}_i)$ depends by Theorem 4.1 only on $\partial(S^*(v))$ and $\mathcal{K}_i$, both of which are determined by Γ and the accumulated set; the identity of the last step e_i enters only through its membership in $\mathcal{K}_i$. $\square$

Corollary 12.2 (Direction without an added postulate). *The relation “ e_{i+1} is available after e_i ” is antisymmetric on record-increasing steps: if committing e makes f available and committing f makes e available, then $e = f$.*

Proof. Suppose $e \neq f$ and each makes the other available. Availability of f after e requires $f \notin \mathcal{K} \cup \{e\}$, and availability of e after f requires $e \notin \mathcal{K} \cup \{f\}$. Committing e then f yields $\mathcal{K} \cup \{e, f\}$ with record $M + 2$; by Theorem 3.4 no operation returns to $\mathcal{K}$, so the second availability is never realised from the first state. Hence $e = f$. $\square$

Remark 12.3 (No further asymmetry is imposed). Theorem 12.2 is obtained from Theorem 3.4 alone. Nothing in this paper adds an orientation to the graph, and the edges of Theorem 2.1 are undirected throughout: the asymmetry is a property of accumulation, not of the edges.

12.1 Constructions that are and are not available

Proposition 12.4 (Deletion of a committed step is not defined). *There is no operation on traces that removes a committed edge and returns a trace of the original graph with a smaller record.*

Proof. Theorem 3.4, final clause. $\square$

Proposition 12.5 (An item is not separable from its complement). *Let U be an item set determined as in Theorem 2.8. There is no determination of U that leaves $\mathbb{C}U$ fixed while removing U , since $\mathbb{C}U$ is defined as $V \setminus U$ and its specification therefore changes with U .*

Proof. By Theorem 2.8 the determination of U is by complementation, and complementation is taken with respect to V . Removing U from V

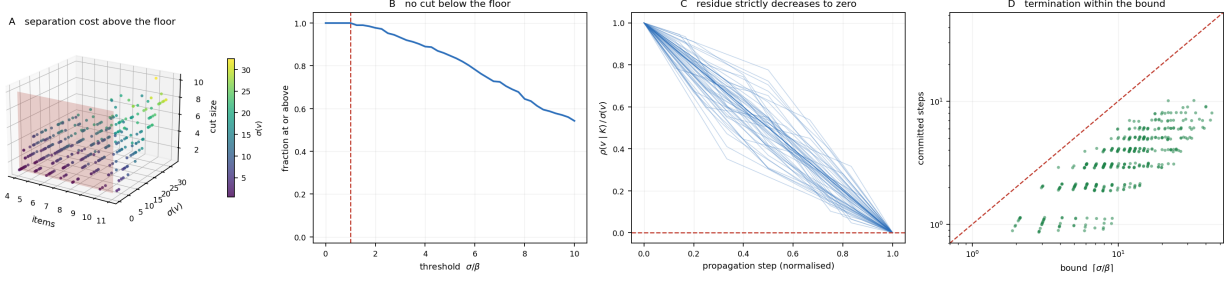


Figure 1: The floor, and residue as the driver of propagation. (A) Three-dimensional scatter of 400 random contact graphs in (number of items, separation cost $\sigma(v)$, minimum-cut size), coloured by $\sigma(v)$ (viridis), with the floor plane drawn in red at the smallest β used. Every point lies on or above the plane: no item is separated from the medium for less than β (Theorem 2.6), and the minimum cut is never empty (Theorem 2.7). (B) Survival curve of the ratio $\sigma(v)/\beta$ across the same graphs. The fraction at or above 1 (red dashed) is exactly unity: no separation falls below the floor in any instance. (C) Residue descent. Each faint curve is one run of the propagation operator $\mathbf{P}$ on one item, with the residue $\rho(v | \mathcal{K})$ normalised by its initial value $\sigma(v)$ and the step index normalised to $[0, 1]$. Every curve is strictly decreasing and every curve terminates at zero (red dashed): a record-increasing commitment exists exactly while residue remains (Theorem 4.3) and each such commitment removes the weight of its edge, hence at least β (Theorem 4.5). (D) Committed steps against the bound $\lceil \sigma(v)/\beta \rceil$ of Theorem 4.6, log-log, jittered. All 400 points lie strictly below the diagonal (red dashed): every propagation terminated, and did so within the predicted number of steps.

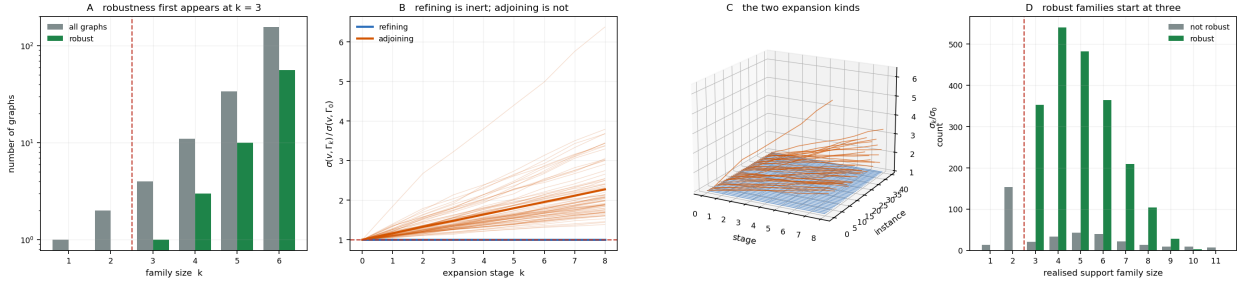


Figure 2: Sufficiency at three, and the asymmetry between refining and adjoining expansion. (A) Exhaustive census of robustness over all graphs on k vertices, $k = 1, \dots, 6$, taken from the graph atlas (grey: all graphs; green: robust ones; log scale). The counts are 0/1, 0/2, 1/4, 3/11, 10/34, 56/156. No graph on one or two vertices is robust and exactly one on three vertices is — the triangle — so the minimal cardinality at which robustness becomes available is three (Theorem 7.5, Theorem 7.6); the red line marks that threshold. (B) Stagewise separation cost $\sigma(v, \Gamma_k)$ normalised by its stage-zero value, along 80 refining families (blue) and 80 adjoining families (orange), eight stages each; heavy lines are the means. The refining mean is 1.000000 at every one of the nine stages — refining medium-attaching expansion leaves the cost exactly fixed (Theorem 8.2) — while the adjoining mean rises monotonically to 2.27 (Theorem 8.4). (C) The same two families in three dimensions, one curve per instance, showing that the refining sheet is flat across every instance rather than flat only on average. (D) Realised support families of 2446 items, by size, split into robust (green) and not (grey). No robust family has fewer than three members; families of three or more may or may not be robust, since robustness is a property of the support graph’s shape and not of its size alone (Theorem 7.7).

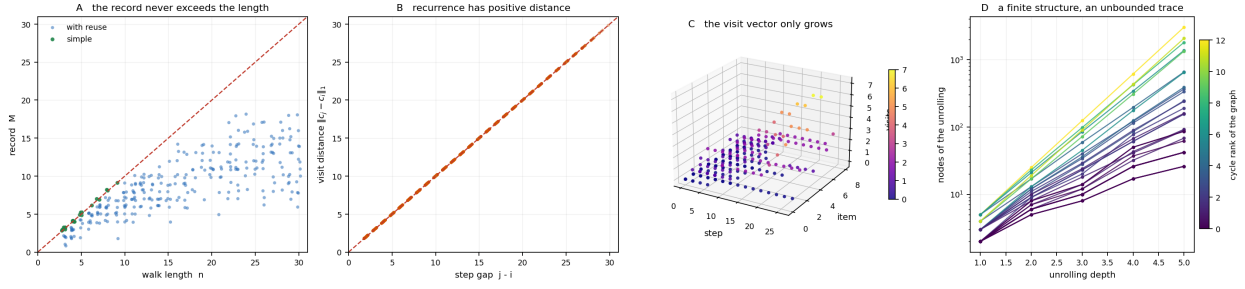


Figure 3: The trace: record, visit vector, recurrence, unrolling. (A) Committed record M against walk length n for 300 random walks of varying length, jittered. Every point lies on or below the diagonal (red dashed), and the points *on* it are exactly the 23 walks that repeated no edge (green): $M \leq n$ with equality if and only if the trace is simple (Theorem 9.3). (B) At each of the 3052 recurrences observed — a walk returning to an item it had already occupied — the visit distance $\|c_j - c_i\|_1$ between the two initial segments plotted against the step gap $j - i$. Every point lies exactly on the diagonal and no point lies at zero: a returning walk agrees with the earlier one in its terminus and differs in its visit vector, so resemblance is never identity (Theorem 9.5, Theorem 9.6). (C) Per-item visit counts along three walks, as (step, item, count) coloured by count. The surface only ever rises: components of the visit vector are monotone and never decrease (Theorem 9.2). (D) Node count of the unrolling against unrolling depth (log scale) for 30 finite contact graphs, coloured by the cycle rank $m - n + 1$ of the graph unrolled. Growth is exponential in depth and ordered by cycle rank, while the graph itself gains no items: a finite structure supports an unbounded monotone trace, the growth being in the traversal and not in the structure (Theorem 9.8, Theorem 9.9). At depth five the unrollings range from 26 to 3019 nodes, and a single item is carried by as many as 103 distinct nodes of the tree.

replaces the ambient set by $V \setminus U$, with respect to which the complement of U is empty. Hence $\mathbb{C}U$ does not survive the removal unchanged. $\square$

Proposition 12.6 (Common support is not a relation between the supported). *Let a, b be items each adjacent to an item c , with no edge $\{a, b\}$. Then no cut separating a from m contains an edge incident to b other than through c or m , and the covariation of the separation costs of a and b under expansions at c is a property of c .*

Proof. Absent $\{a, b\}$, the only paths from a to b pass through c or m ; so the edges available to a minimum a – m cut that also touch b are those incident to c or m . An expansion at c alters cuts through c and hence may alter both costs; the alteration is computed from c and its incident weights. $\square$

Remark 12.7 (What the three propositions rule out). Theorem 12.4 rules out an operation returning a trace to an earlier record. Theorem 12.5 rules out varying an item while holding its complement fixed. Theorem 12.6 locates covariation of two separations in the item they share rather

than in a relation between them. Each is a statement about finite weighted graphs and walks on them; no theorem below depends on any further reading.

13 Scope

13.1 What is claimed

The results are theorems about finite weighted graphs and finite walks on them. Specifically: that residue as defined in Theorem 4.1 is positive exactly when a record-increasing commitment remains (Theorem 4.3); that it splits uniquely into a consumed and an invariant part (Theorem 5.4); that propagation halts exactly at a fixed point of Theorem 6.1, which is exactly the identity process (Theorem 6.2); that the object standing there is intrinsically characterised and requires no stipulated resting configuration (Theorem 6.5); that its separation is carried by a robust mutually supporting family of minimal size three (Theorem 7.6); that refining expansions leave the intrinsic floor fixed while adjoining ones raise support until closure (Theorem 8.2, Theorem 8.4, Theorem 8.6); that traces carry a visit vector

strictly finer than the record, monotone componentwise, giving recurrence positive distance (Theorem 9.3, Theorem 9.5); and that hoisting is not level-respecting and must complete before resolution (Theorem 10.6, Theorem 10.7).

Section 11 reports a randomized verification of all of these on 300 concrete contact graphs, 205,536 checks, no failures.

13.2 What is not claimed

We do not claim that the intrinsic floor of Theorem 5.1 exists for items not present in every stage of the expanding family; the limit is taken over stages containing the item, and nothing is asserted about stages that do not.

We do not claim that every expansion leaves β^* fixed. Theorem 8.2 is stated for refining medium-attaching expansions, and Theorem 8.3 records that expansions failing either hypothesis may move the intrinsic floor. Such a move is a change of the item under Theorem 5.4, and we do not treat the question of when two stages should be said to contain the same item.

We do not claim that the number three in Theorem 7.6 is the size of any family occurring in a particular application; the theorem gives the minimal cardinality of a robust mutually supporting family, and a given separation may be carried by more.

We do not claim that Theorem 7.3 is the only reasonable notion of independence. It is the one under which Theorem 8.2 and Theorem 8.4 are both provable, and we have not investigated alternatives.

We do not derive a rate: nothing here says how many stages are required for an item to close, only that closure coincides with the fixed point (Theorem 8.6).

13.3 Relation to existing work

The floor derivation of Theorem 2.6 is a statement about minimum cuts in the sense of Ford and Fulkerson (1956) and Menger (1927), and the computability remarks throughout rely on standard maximum-flow algorithms (Ahuja et al., 1993; Goldberg and Tarjan, 1988). The unrolling of Theorem 9.7 is the standard universal cover construction of a graph read as a tree of walks. The monotone-accumulation results of Section 3 are elementary and are included because later sections use their exact form. We are not aware

of a treatment in which the terminus of a separating process is characterised as a fixed point of the operator that performs it, and this together with Theorem 7.6 and Theorem 8.2 is what we take to be new.

14 Conclusion

One mechanism has carried the paper. A separation leaves a remainder; the remainder is what makes a further separation possible; and a process is the consumption of that remainder. Everything else follows by taking the mechanism seriously.

Taken seriously, it says first that a process cannot run past the point at which its reducible remainder is exhausted, and that this point is not an abandonment but a fixed point of the operator that performs the process — the unique place where the operation is the identity and its output equals its input (Theorem 6.2). An entry of a catalogue of resting items sits there, and sits there intrinsically (Theorem 6.5).

It says second that the separation standing at that point is never carried by one contact. It is carried by a family that survives the loss of any of its members, and the smallest such family has three members because a triangle survives a deletion that a path does not (Theorem 7.6). This makes sufficiency transverse: refining one line of separation is inert on the intrinsic floor (Theorem 8.2), while adjoining independent lines raises support until closure, after which further lines resolve to a separation already reached (Theorem 8.6). The resulting invariance holds in both indices of the expanding family (Theorem 8.7).

It says third that the record of what has been committed admits no decrement, and that the finer invariant of a trace — its visit vector — gives recurrence a positive distance, so that a walk returning to an item it has occupied is not returned to an earlier state (Theorem 9.5). A finite structure with a reachable cycle therefore supports an unbounded monotone trace, the growth being in the traversal and not in the structure (Theorem 9.9).

And it says finally that an operation placing something globally from a local position must be settled before anything reads it (Theorem 10.8), and that nothing appears in an output which was not witnessed by an actual traversal of the input (Theorem 10.3) — which is, by Theorem 10.4,

the monotonicity of the record again, in a third costume.

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